W08

Species occurrence data from GBIF

R PROGRAMMING GIS & MAPPING

GBIF (the Global Biodiversity Information Facility) holds hundreds of millions of species records from museums, citizen-science projects and surveys. They are very valuable but come with errors and biases. Today you learn to download them, clean them with explicit criteria and detect sampling bias.

DURATION	About 2 hours.
PACKAGES	`rgbif`, `CoordinateCleaner`, `sf`, `tidyverse`
FILES	`week08_GBIF_data.R`, `week08_solutions.R`, `gbif_lullula_arborea_raw.csv`, `montfred.gpkg`
DATA	286 woodlark (Lullula arborea) records imitating a GBIF download, with typical errors added on purpose.

Learning outcomes

- search and download records with `rgbif`, and know when to use `occ_download()`;
- recognise the most common errors in occurrence records;
- clean data step by step and with `CoordinateCleaner`, documenting each step;
- detect sampling bias and reduce it with spatial thinning.

1 · Downloading from GBIF

```
library(rgbif)
key <- name_backbone("Lullula arborea")$usageKey      # taxonomic identifier
occ_search(taxonKey = key, country = "ES", hasCoordinate = TRUE,
           year = "2015,2025", limit = 500)$data        # for exploring
# For a paper: occ_download() with your GBIF account; it gives a DOI you must cite
```

In class we use a prepared file so everyone has the same data and the same errors. In the script the real download sits inside `if (run_gbif)`: set `run_gbif` to TRUE to try it.

2 · Typical errors

Problem	How to detect it
Coordinates at (0, 0)	CoordinateCleaner "zeros" test
Swapped latitude and longitude	the point lands on another continent; spatial filter
Country capital or centroid (default georeferencing)	"capitals" and "centroids" tests
Uncertainty of several km	`coordinateUncertaintyInMeters`
Fossils and old museum specimens	`basisOfRecord, year`
Duplicates (same record on two platforms)	`distinct()`

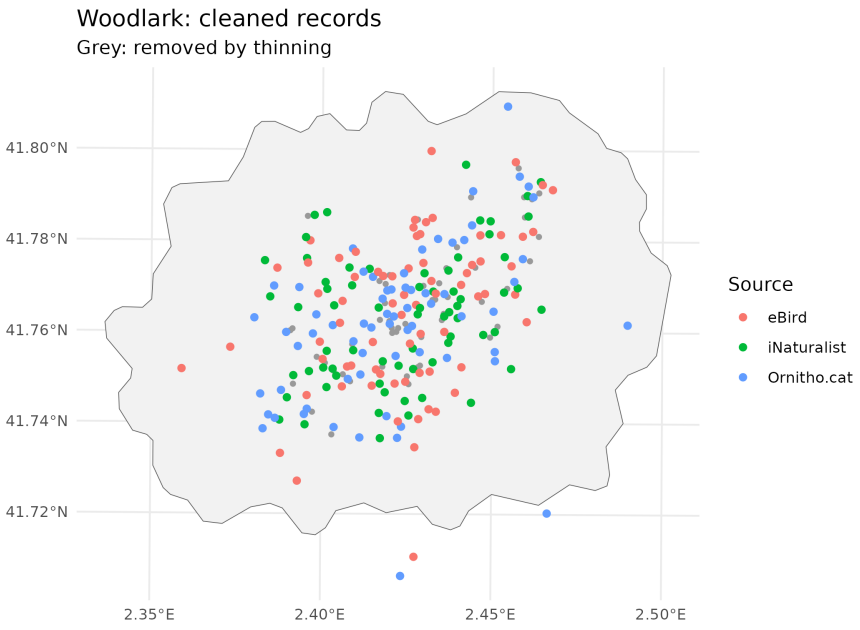
3 · Clean and document

The script records how many records remain after each step with a small function, `log_step()`. That table goes straight into your methods section:

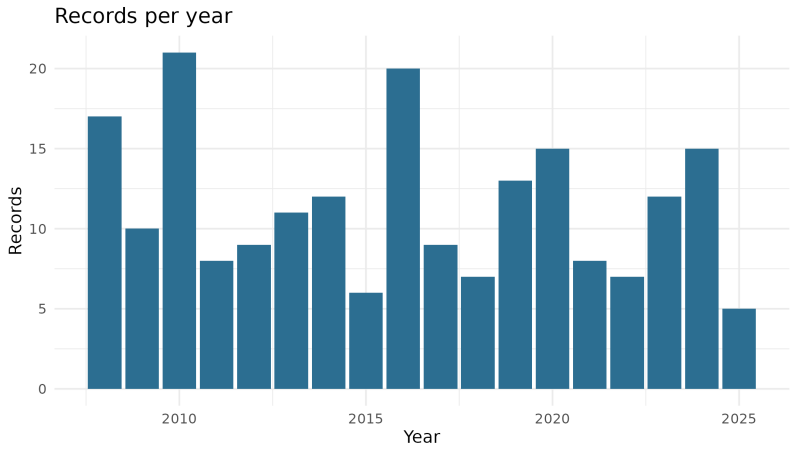
Step	Remaining
0 Raw download	286
1 Observations only (no fossils or specimens)	284
2 Year 2000 or later	282
3 Uncertainty 1,000 m or less	279
4 No duplicates	267
5 CoordinateCleaner	263
6 Inside the study area	260
7 Thinned to one per 250 m cell	205

4 · Sampling bias and thinning

Records pile up where people go: paths, viewpoints, huts. A model that learns from those clusters learns "where observers are". Keeping one record per 250 m cell reduces that effect. Also look at which sources contribute most data and in which years.



Cleaned records by source. Grey: removed by thinning.



Records per year after cleaning.

Cite the data

Downloads made with occ_download() have a DOI. Cite it in any work using those data: it is GBIF's condition of use and credits the people who collected them.

Exercises

Answers in `week08_solutions.R`.

1. Table of how many records each cleaning step removed.
2. Which CoordinateCleaner tests flagged records?
3. Repeat thinning with 500 m cells. How many records remain?
4. Which source contributes most records? Could it bias a model?
5. With internet: download woodlark records for your own region with rgbif.

Before Week 9

- Install `install.packages(c("spdep", "gstat", "spatstat"))`.
- Keep `gbif_lullula_arborea_clean.csv`: we use it in Week 10.